



POLITECNICO
MILANO 1863

Node-RED

Evaluation Lab

Luca Mottola

luca.mottola@polimi.it

<http://mottola.faculty.polimi.it>

Sensor Data Monitor

- Sensed data originating from different sources is not necessarily coherent
- You are to implement a Node-RED flow that checks whether OpenWeatherMap and **sensor.community** produce coherent data
 - Compute the average temperature from MQTT over **neslabpolimi/smartcity/milan** over a sliding window of ten consecutive readings
 - Query OpenWeatherMap once every minute
 - Check whether the two differ for more than a value **K**
- If the data is not coherent, publish “Temperature data is not coherent!” over **neslabpolimi/nsds/eval24/alarm**

Notes

- The value **K** comes from MQTT over **neslabpolimi/nsds/eval24/k**
 - It is periodically published there
 - It may change over time!
- You are to compute the **average temperature also for incomplete windows**
- You can use Celsius or Kelvin measurements
- Be careful with context: choose the appropriate scope!

Rules

- Complete the README.md file with
 - Your group identifier
 - From the group registration document
 - Name of each group member
 - A 200-word (max) description of the message flows in your solution
 - What node sends what message to what other node, when, ...
 - List of extensions possibly used besides the ones seen during the regular labs
- Submit a single zip file with **only** the flow file and README.md
 - Remove everything that is not the project you want to submit
 - Other flows in different tabs, notes, ...
 - Make sure to export **the whole flow!**
 - Name of the file: nodered-groupXX.zip
 - XX is the group identifier from the group registration document
 - Submit by the user corresponding to the contact email specified in the group registration document
- Failure to stick to these rules yields a deduction of one point